THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

May 06, 2019 Time: 2:30 pm - 5:30 pm

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Properties of water:

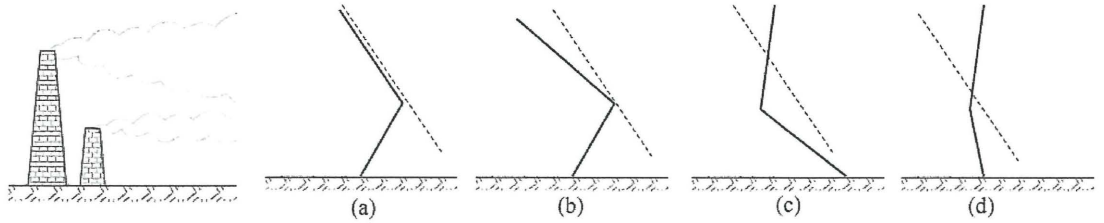
Density= 1000 kg/m3
Viscosity (15 °C) = 1.1404 mPa· s
Specific heat= 4.18 kJ/(kg. 0 C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion= 333 kJ/kg

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. Single choice questions (30 marks, 3 marks for each)

1. A 1000-Mw_e coal-fired power plant in Hong Kong has an efficiency of 33.3% with 15% of the waste heat being released to the atmosphere as stack heat and the other 85% taken away in the cooling water. If the cooling water is allowed a rise in temperature by 8.0 ^{0}C, what flow rate would be required, in m^3/s: (a) 40.8, (b) 50.8, (c) 60.8, or (d) 80.8.
2. The same 1000-Mw_e coal-fired power plant, without air pollution control, would generate 216 t/d of sulphur emissions. Assuming 85% efficiency of the sulphur emission control system, how many cubic kilometres of clean air volume will be required daily for dilution in order to meet the air quality objective of 80 microgrammes per cubic metre sulphur dioxide: (a) 8.1, (b) 81, (c) 810, or (d) 8100.
3. Which one of the following descriptions of a Hong Kong municipal solid waste landfill is false: (a) landfill gas consists mostly of methane and carbon dioxide, (b) the generated landfill gas may be used to produce electricity, (c) landfill leachate may emit methane to the atmosphere and contribute to global warming, and (d) landfilling municipal solid waste requires more space than a municipal solid waste incineration plant.
4. Hydrogen sulfide (H_2S) is an odorous toxic gas, which dissolves in water forming H+ and HS^-. The equilibrium constant for this reaction is 0.86×10^{-7}. In order to prevent H_2S formation, the pH of the water has to be adjusted. At which of the following pH levels is the fraction of hydrogen sulfide in H_2S form less than 1%: (a) 5.1, (b) 6.1, (c) 8.1, or (d) 9.1.
5. Calcium (Ca^{2+}), magnesium (Mg^{2+}) and sodium (Na^+) are among the most common cations found in groundwater. Which of the following waters has the highest degree of hardness as expressed by $CaCO_3$ equivalent? Water (a) containing 60 mg/L of Ca^{2+}, 24.3 mg/L of Mg^{2+} and 23 mg/L of Na^+, or (b) containing 40 mg/L of Ca^{2+} and 48.6 mg/L of Mg^{2+}, or (c) containing 97.2 mg/L of Mg^{2+}, or (d) containing 80 mg/L of Ca^{2+} and 69 mg/L of Na^+. (Atomic weights are: Ca 40; Mg 24.3; and Na 23.)
6. The purpose of water treatment systems is to bring raw water up to drinking water quality. Generally, different treatment procedures are used to treat surface water and groundwater. Which one of the following statements is false: A typical surface water treatment may include the following treatment processes: (a) flocculation, (b) recarbonation, (c) sedimentation, and (d) disinfection.
7. The effects of air pollution has been directed toward six criteria pollutants. Which one below is not one of the six criteria pollutants? (a) N_2O, (b) O_3, (c) CO, (d) NO.
8. A tall stack and a nearby short stack have plumes as shown below. Which atmospheric temperature profile would be most likely to cause that pair of plumes?



9. Two identical membrane systems are connected in series. If each membrane is able to achieve a 2-log removal of target contaminant. Which is the overall rejection rate of the contaminant? (a) 90%, (b) 99%, (c) 99.9%, (d) 99.99%.
10. Ngong Ping Sewage Treatment plant is the first tertiary treatment work in Hong Kong in order to protect the water quality of the water gathering ground and other receiving water bodies. What is the main purpose of the tertiary treatment? (a) to remove suspended particles, (b) to remove BOD, (c) to remove sludge, (d) to remove nutrients.

Section B (4 questions with unequal marks)

11. Briefly describe the following technical terms and discuss their respective applications in environmental engineering: **(20 marks, 4 marks** for each one)

    a) Adiabatic lapse rate
    b) Sedimentation
    c) Sanitary landfill
    d) Global Warming Potential (GWP)
    e) Water hardness and alkalinity

12. A wastewater treatment plant discharges 1.0 m^3/s of effluent having a 5-day BOD_5 of 30 mg/L into a stream flowing at 10.0 m^3/s. Just upstream from the discharge point, the stream has an ultimate BOD of 3.0 mg/L. Both BOD reaction rate constant k and deoxygenation constant k_d have been found to be 0.22/day.

    (a) What would be the ultimate BOD for the wastewater? **(4 marks)**
    (b) Assuming complete and instantaneous mixing, find the ultimate BOD of the mixture of waste and river just downstream from the outfall. **(4 marks)**
    (c) Assuming a constant cross-sectional area for the stream equal to 55 m^2, what ultimate BOD would you expect to find at a point 20,000 m downstream? **(4 marks)**
    (d) Describe the changing trend of organic waste amount, BOD and DO value from outfall to a long distance of downstream. **(5 marks)**

13. IPCC Special Report on Emission Scenarios is based on a set four storylines that weave demographic, technological, and economic driving forces into four distance scenarios that can be translated into future emissions. One of the defined scenarios, in China and centrally planned Aisa, averaged annual growth rate of population is 1.03%, GDP per capita growth rate is 3.91%, growth rate of energy intensity is -1.73% and carbon intensity growth rate is -0.32%. Assume the atmospheric fraction constant at 0.38. Climate sensitivity parameter ΔT_{2X} for a doubling of CO_2 is 3°C.

    In 2019, the initial CO_2 concentration is 400 ppm, the energy-related emission rate is 8.0 GtC/yr, and the initial atmospheric carbon content is 848 GtC. Land-use emission rate is a constant 5.5 GtC/yr:

    (a) Estimate the carbon emission rate in 2079 and total carbon emission from 2019 to 2079. **(5 marks)**
    (b) What would be CO_2 concentration in 2079? **(3 marks)**
    (c) Estimate CO_2-induced global equilibrium temperature increase in 2079 **(4 marks)**
    (d) If those growth rates continue, in what year would CO_2 concentrations be equal to double the initial 400 ppm? **(4 marks)**

14. A confined aquifer with a thickness of B=30.0 m has been pumped from a fully penetrating well with a radius of r_w=0.20m at a steady rate of Q=5000 m^3/day for a

long time. Drawdown at an observation well 15 m from the pumped well is 1.8 m, and drawdown at a second observation well 150 m away is 0.30 m.

(a) Find the hydraulic conductivity of the confined aquifer **(5 marks)**

(b) Estimate the drawdown at the pumped well **(4 marks)**

(c) Generally, the length of time required for groundwater to flow from an observation well to a pumped well in a confined aquifer can be expressed by $t = \frac{\pi B \eta (r^2 - r_w^2)}{Q}$, where η means the porosity = 0.30, r means the radial distance from the observational well to the pumped well. If the observational well 150m away is found to be contaminated, to determine how long the contamination can be transported to the pumped well.

(4 marks)

(d) Discuss what measures can we take to control the movement of contamination, and how it works **(4 marks)**

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